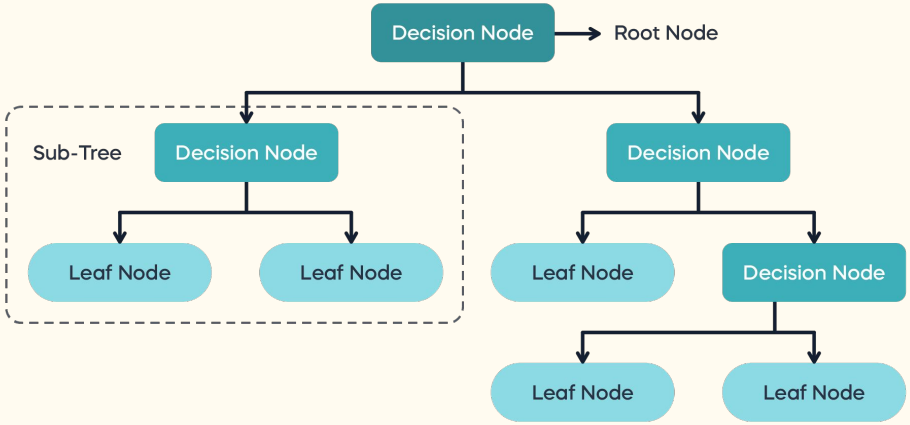
Information in Decision Trees

—

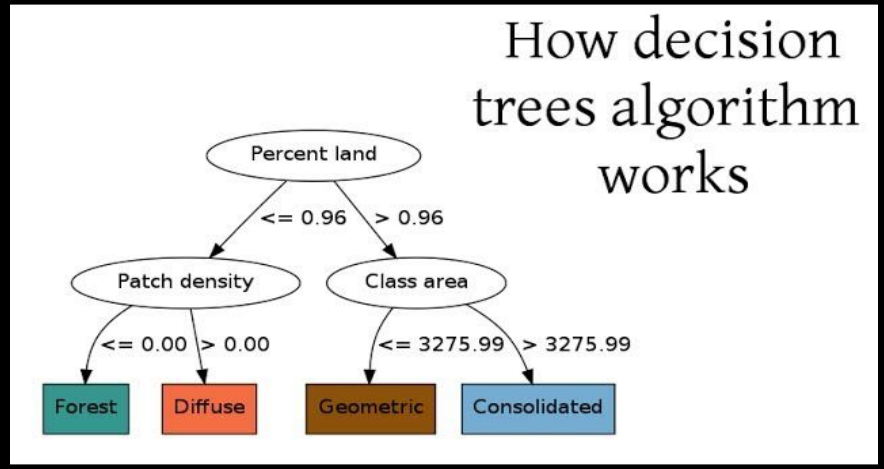
Thanh + Nam

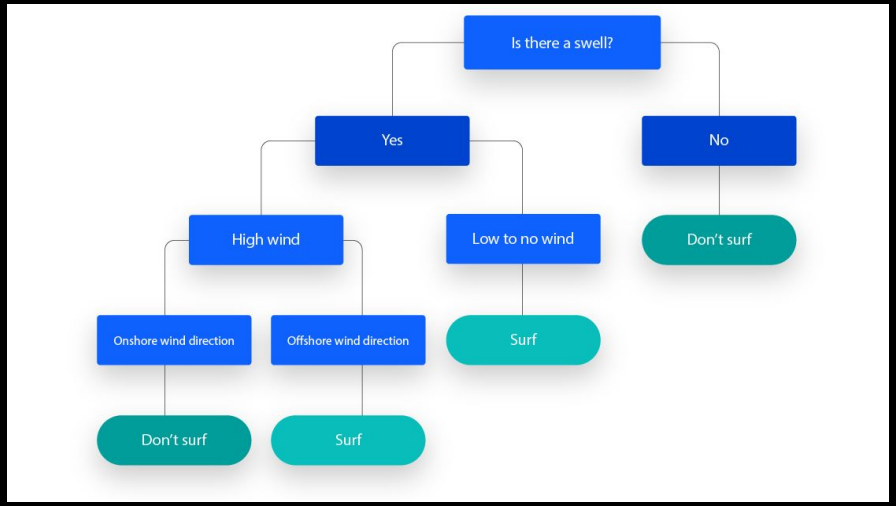
Introduction for decision tree



Structure of a decision tree

How decision trees work?





Advantages and disadvantages of decision tree

1. Simplicity and Interpretability
2. Versatility
3. No Need for Feature Scaling
4. No Need for Feature Scaling

1. Overfitting
2. Instability
3. Bias towards Features with More Levels

Entropy and Information Gain

1. Entropy

Dataset:

Weather	Humidity	Wind	Play Tennis?
Sunny	High	Weak	No
Sunny	High	Strong	No
Overcast	High	Weak	Yes
Rain	High	Weak	Yes
Rain	Normal	Weak	Yes
Rain	Normal	Strong	No

Step 1: Initial Entropy

The dataset has 6 samples:

- 3 "Yes," 3 "No."

$$H(S) = -\left(\frac{3}{6}\log_2\frac{3}{6} + \frac{3}{6}\log_2\frac{3}{6}\right) = 1.0$$

Step 2: Splitting on "Weather"

1. Sunny (2 "No"):

$$H(Sunny) = 0$$

2. Overcast (1 "Yes"):

$$H(Overcast) = 0$$

3. Rain (2 "Yes," 1 "No"):

$$H(Rain) = -\left(\frac{2}{3}\log_2\frac{2}{3} + \frac{1}{3}\log_2\frac{1}{3}\right) \approx 0.918$$

Weighted Entropy:

$$H(S|Weather) = \frac{2}{6}(0) + \frac{1}{6}(0) + \frac{3}{6}(0.918) = 0.459$$

Step 3: Splitting on "Humidity"

1. High (2 "No," 1 "Yes"):

$$H(High) = -\left(\frac{2}{3}\log_2\frac{2}{3} + \frac{1}{3}\log_2\frac{1}{3}\right) \approx 0.918$$

2. Normal (2 "Yes"):

$$H(Normal) = 0$$

Weighted Entropy:

$$H(S|Humidity) = \frac{3}{6}(0.918) + \frac{2}{6}(0) = 0.459$$

2. Information gain

Information Gain:

$$IG(S, Weather) = H(S) - H(S|Weather) = 1.0 - 0.459 = 0.541$$

Information Gain:

$$IG(S, Humidity) = H(S) - H(S|Humidity) = 1.0 - 0.459 = 0.541$$

Entropy and information gain are essential for building decision trees

1.Quantify the quality of splits (entropy).

2.Ensure optimal attribute selection (information gain).

3.Guide the tree to create subsets that are as pure as possible, improving its predictive power.

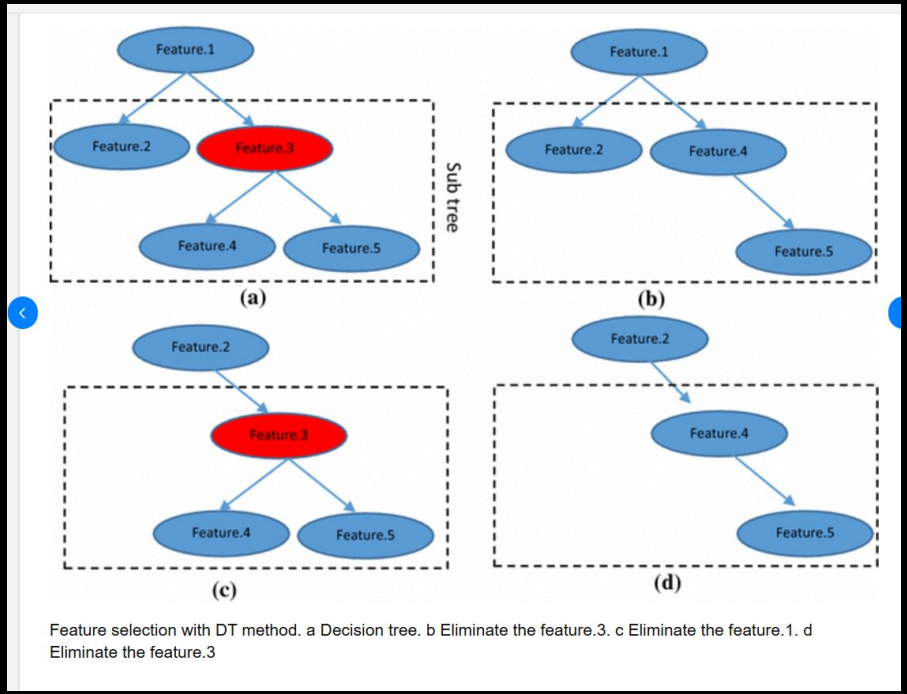
By prioritizing splits with the highest information gain, decision trees achieve accurate and interpretable results efficiently.

Handling Large Datasets

Feature Selection

Feature selection helps in:

1. Increasing model performance.
2. Increasing model interpretability.
3. Reducing model complexity.
4. Enhancing data visualization.



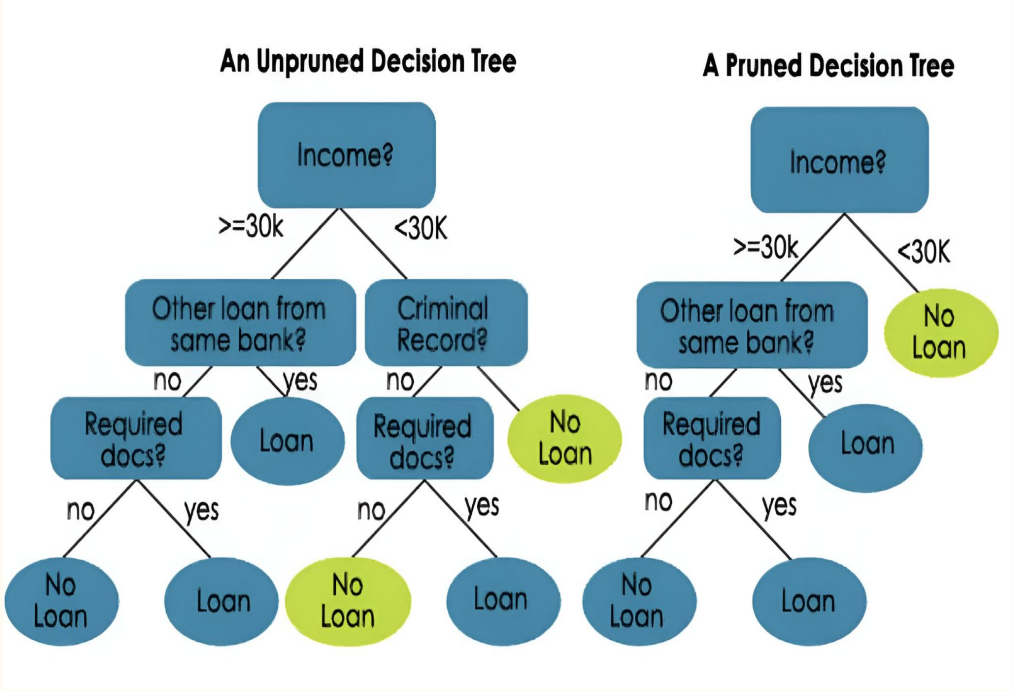
Feature selection with DT method. a Decision tree. b Eliminate the feature.3. c Eliminate the feature.1. d Eliminate the feature.3

Pruning Strategies

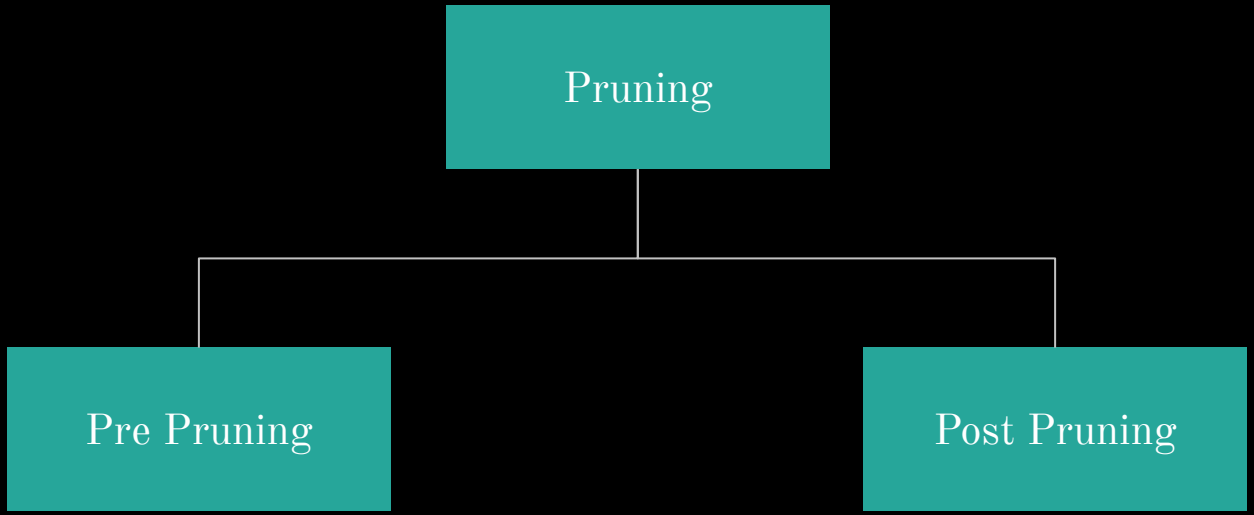
Definition

- The process of reducing branches or nodes from a decision tree
- Only keep informative branches

Improve generalisation
Prevent overfitting



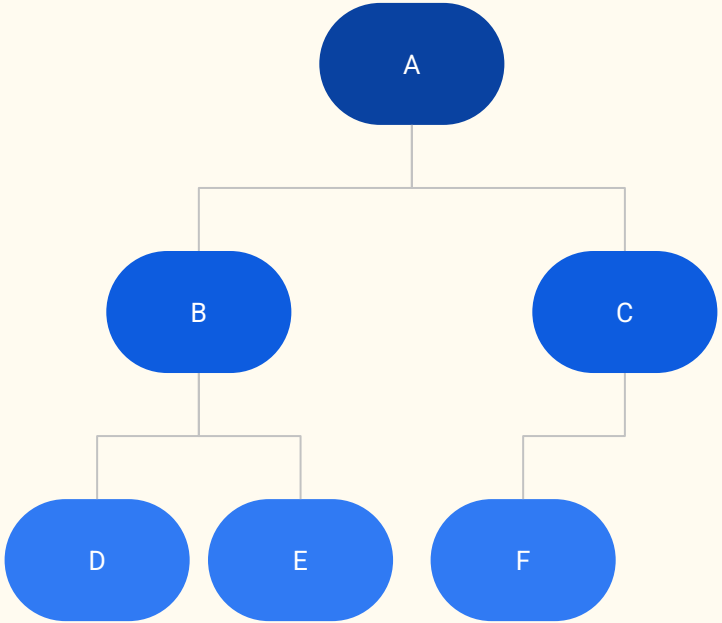
Main Types



Pre Pruning

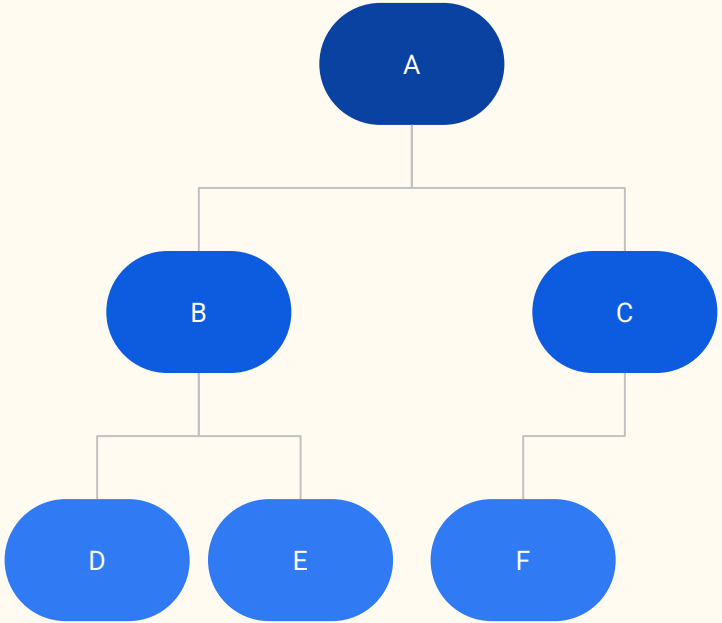
- A tree is “pruned” by halting its construction early when you reach a certain threshold
- 5 common methods to determine the threshold:
  - Maximum depth
  - Minimum samples per leaf
  - Minimum samples per split
  - Maximum features
  - Maximum Leaf Nodes

Maximum depth

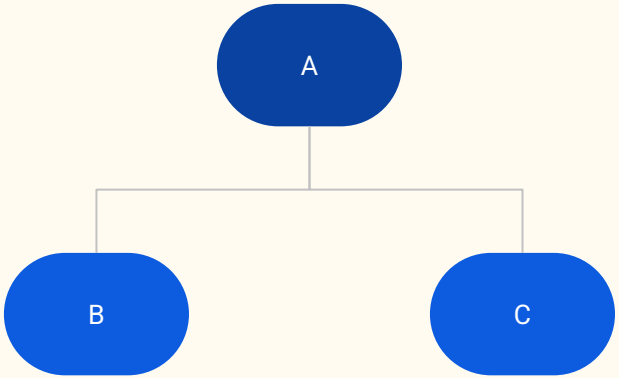


Normally

Maximum depth

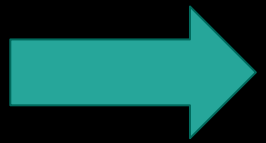


Normally



Set max depth = 2

Post pruning

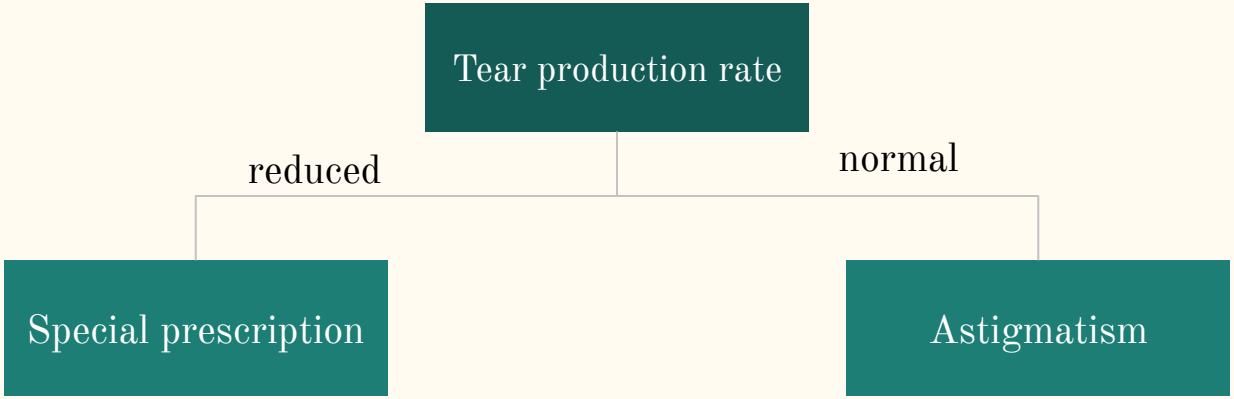


More reliable

Post Pruning

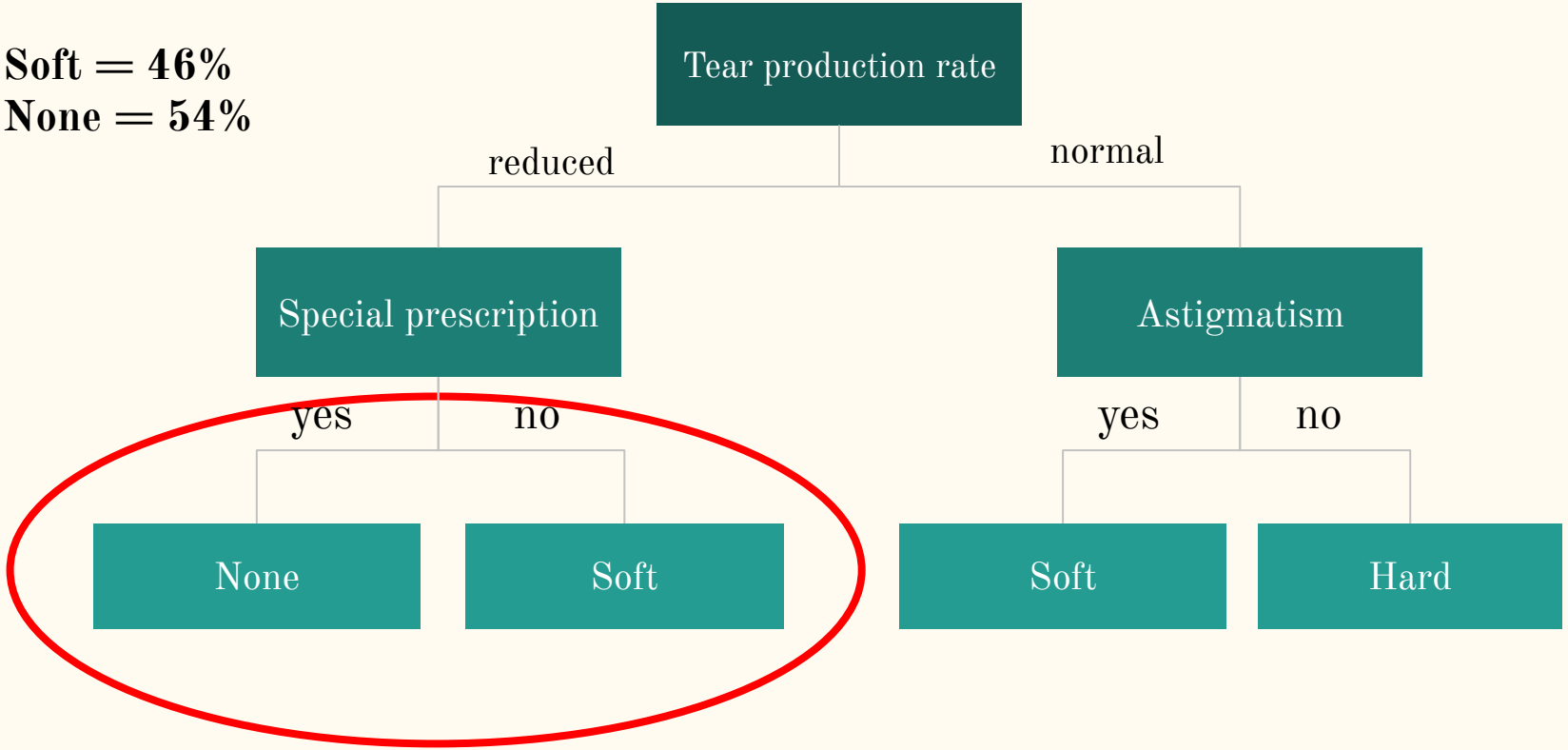
- **Validation Set**: A portion of the training data is typically reserved as a validation set to guide the pruning process.
- **Fully Grown Tree**: Initially, the decision tree is grown to its maximum depth
- **Pruning Process**: Afterward, branches are removed if they do not improve performance
- **Error Complexity Pruning**: Each branch is replaced with a leaf node if pruning reduces the error on a validation dataset or based on a cost-complexity trade-off.

Determine contact lens type

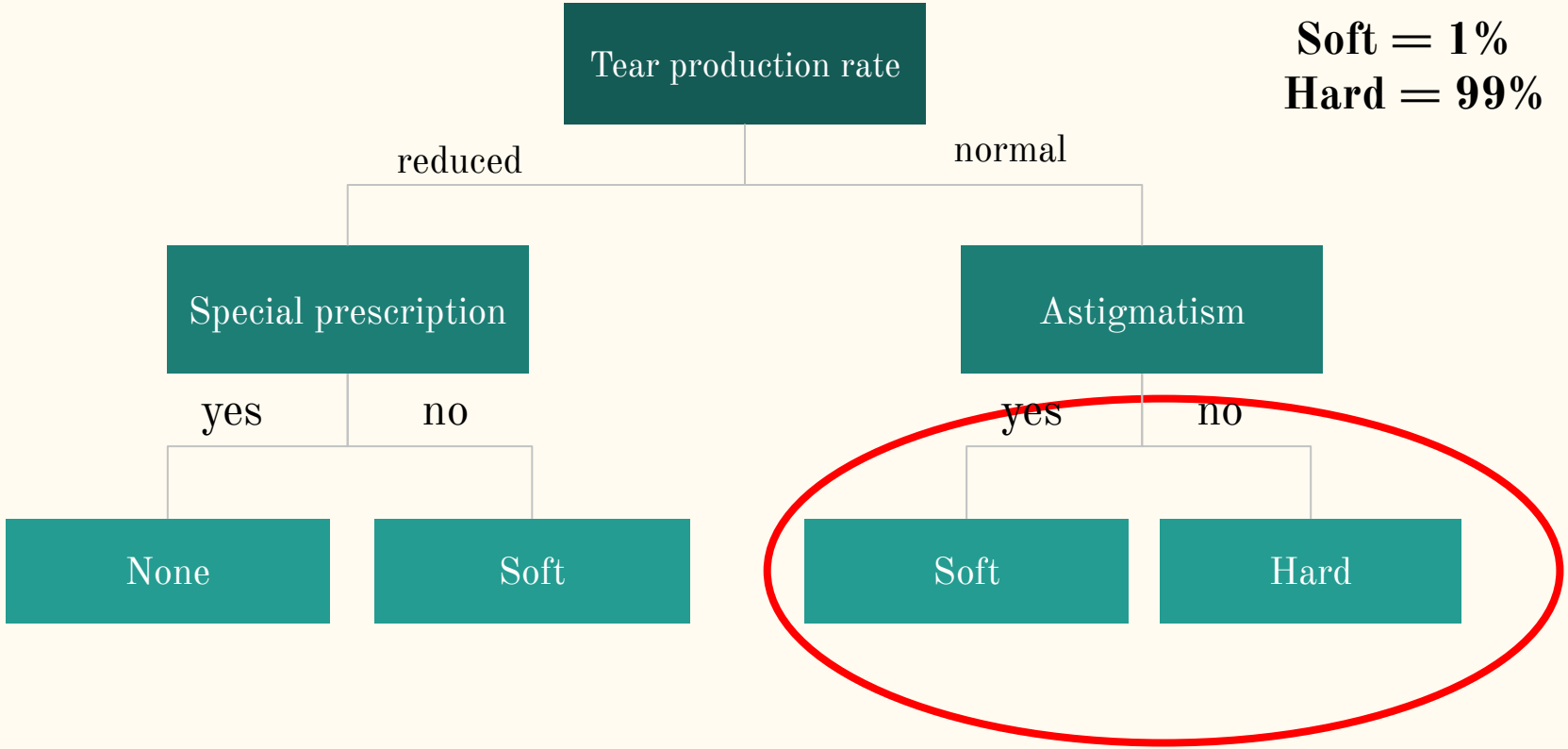


Determine contact lens type

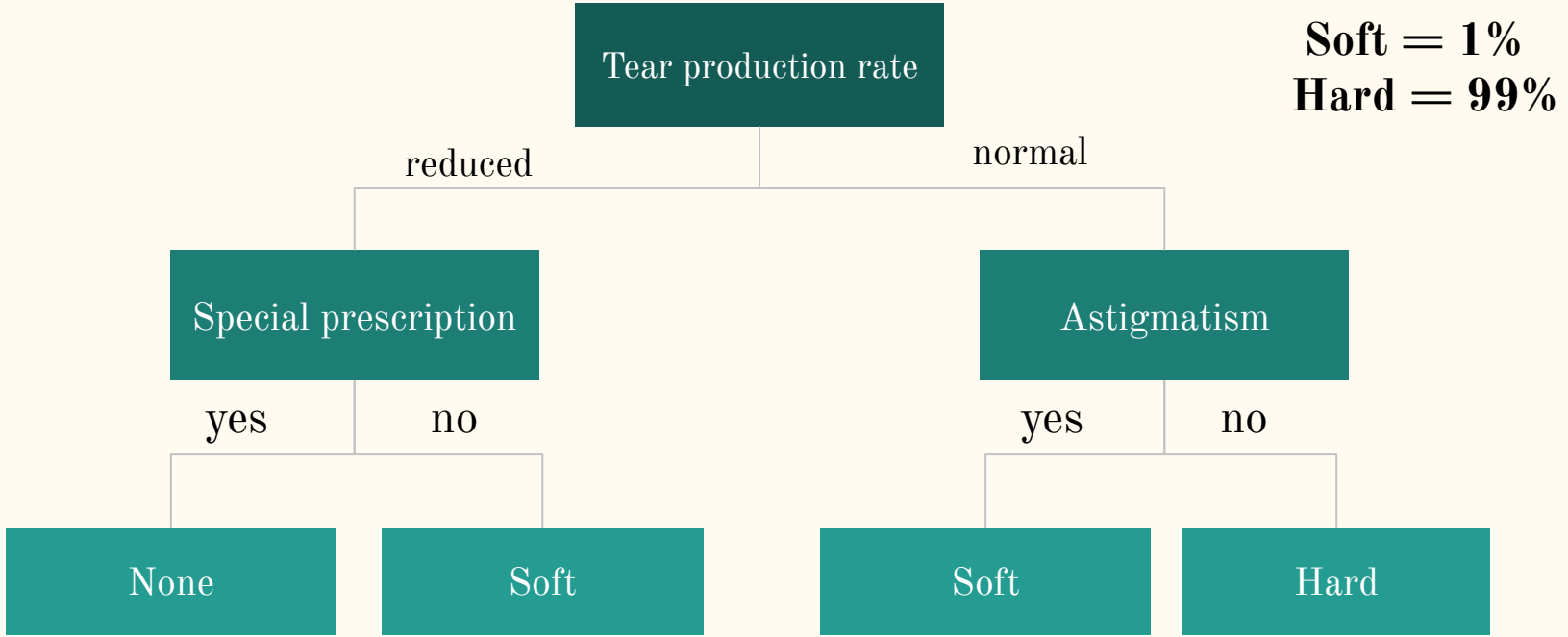
Soft = 46%
None = 54%



Post Pruning Process

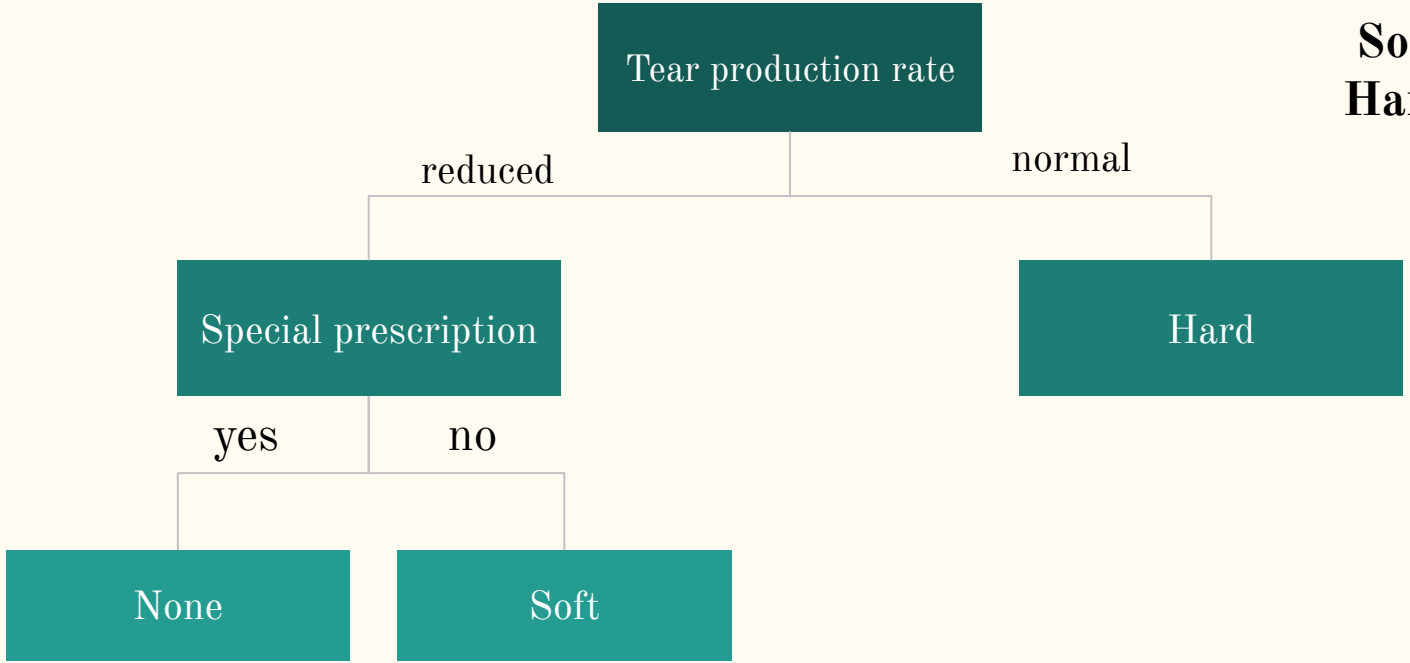


Post Pruning Process



Soft = 1%
Hard = 99%

Post Pruning Process



Soft = 1%
Hard = 99%

Analysis of Post Pruning

Advantages:

- **Better Generalization:** Consider all possible patterns before discarding irrelevant
- **Avoids Overfitting:** Post-pruning typically results in a simpler model that generalizes better to unseen data.

Drawbacks:

- **Slower:** Requires growing the tree fully
- **Validation Set Required:** Requires a validation set to determine which branches to prune

THANK YOU FOR LISTENING